

VIJAY KARANJKAR

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EDUCATION

Master of Science in Computer Science

Graduating May 2026

Arizona State University, Tempe, AZ, USA

Ira Fulton College of Engineering

Relevant coursework: Probabilistic Learning and Reasoning, Knowledge Representation

Bachelor of Technology in Computer Science and Engineering

Graduating May 2024

Indian Institute of Information Technology Vadodara, Gandhinagar, Gujarat, IN

8.64/10 CPI

Gandhinagar Campus

Relevant coursework: Information Retrieval, Artificial Intelligence

TECHNICAL SKILLS

Programming: Python, C, C++, Java, Javascript, Go, MATLAB

Database Systems: SQL, MongoDB

Python Libraries: Matplotlib, SciKit Learn, Pytorch, Lightning.ai, Pandas, Numpy

Miscellaneous: RabbitMQ, Express.js, Node.js, React

PROFESSIONAL EXPERIENCE

Bhabha Atomic Research Center, Mumbai, Maharashtra, IN: Project Intern

February 2024 - May 2024

- Engineered methods for simultaneous inference for classification tasks related to face recognition, emotion and spoofing detection by training and testing multiple models on a database of 1.2 million images.
- Formalized techniques for synthetic in-context data generation (LLM and fairsequence) and implemented machine translation system with a BLEU4 score of 22.

Tikaj Security, Remote Workplace: SDE Intern

September 2023 - December 2023

- Innovated a web crawler system for instant notifications and continuous monitoring of over 1,000 websites, leveraging MongoDB and RabbitMQ for a scalable system with efficient sequential data flow.
- Implemented a caching mechanism using Redis to optimize data retrieval, reducing latency to 0.2 seconds and improving overall system performance.

Extent, Remote Workplace: Frontend Developer Intern

December 2022 - January 2023

- Designed and developed six pixel-perfect dashboards for visualizing complex user data, collaborating closely with design team to ensure a seamless and consistent UI/UX.
- Created a highly responsive website using React, TypeScript, and SASS, boosting user engagement metrics by improving UI/UX; streamlined development processes, enabling faster updates and reducing bug resolution time by 10%.

PROJECTS

I-Smart Research Scholar

October 2022

Managed a team of five to develop an application for performing analysis and summarizing research papers.

- Programmed a website to search and analyse research papers within 3 seconds.
- Applied concepts of natural language processing and information retrieval to generate summary and insights on research papers while user query relevance was measured using bag of words model.
- Presented to 4 jury members who criticised and guided throughout the hackathon.
- Recognized by Government of Gujarat Student Startup and Innovation program and was awarded first prize all over Gujarat with 461 participating teams.

OTHER WORK EXPERIENCE

Indian Institute of Information Technology Vadodara, Gujarat, IN: Research Intern

May 2023 - July 2023

- Collaborated with a professor to work on a research project related to LLMs, Multimodal RAG and Multi-Agent system.